

Inverse By Rows Elementary Operation

Q1: $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$

Sol:

$[A : I]$

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}, I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A:I = \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array} \right] \begin{array}{l} \\ R_2 - 2R_1 \\ R_3 - R_1 \end{array}$$

$$= \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right] \begin{array}{l} \\ R_3 - 2R_2 \\ -R_3 \end{array}$$

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & 9 & 1 & 5 & -2 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right] \begin{array}{l} \\ R_1 - 2R_2 \\ \end{array}$$

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -40 & 16 & 9 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right] \begin{array}{l} R_1 - 9R_3 \\ R_2 + 3R_3 \end{array}$$

$$A^{-1} = \begin{bmatrix} -40 & 16 & 7 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}$$

$$(A^{-1} \cdot A) = \begin{bmatrix} -40 & 16 & 7 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$$

$$A^{-1} \cdot A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

hence proved

$$A \cdot A^{-1} = I$$

Ans

Q2: $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$

Sol:

$[A : I] = A^{-1}$

$$A^{-1} = \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 4 & 5 & 0 & 1 & 0 \\ 3 & 5 & 6 & 0 & 0 & 1 \end{array} \right] \begin{array}{l} \\ R_2 - 2R_1 \\ R_3 - 3R_1 \end{array}$$

$$= \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 0 & -1 & -2 & 1 & 0 \\ 0 & -1 & -3 & -3 & 0 & 1 \end{array} \right] \begin{array}{l} \\ -R_3 \leftrightarrow R_2 \\ \end{array}$$

$$= \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 3 & 3 & 0 & -1 \\ 0 & 0 & -1 & -2 & 1 & 0 \end{array} \right] \begin{array}{l} \\ -R_3 \\ \end{array}$$

$$= \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & 3 & 3 & 0 & -1 \\ 0 & 0 & 1 & 2 & -1 & 0 \end{array} \right] \begin{array}{l} \\ R_1 - 2R_2 \\ \end{array}$$

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & -3 & -5 & 0 & 2 \\ 0 & 1 & 3 & 3 & 0 & -1 \\ 0 & 0 & 1 & 2 & -1 & 0 \end{array} \right] \begin{array}{l} \\ R_2 - 3R_1 \\ R_1 + 3R_2 \end{array}$$

$$A^{-1} = \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -3 & 2 \\ 0 & 1 & 0 & -3 & 3 & -1 \\ 0 & 0 & 1 & 2 & -1 & 0 \end{array} \right] \Rightarrow A^{-1} = \begin{bmatrix} 1 & -3 & 2 \\ -3 & 3 & -1 \\ 2 & -1 & 0 \end{bmatrix}$$

Solved

Q3: $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

Sol: $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$

$A^{-1} = [A : I]$

$A^{-1} = \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 3 & 4 & 0 & 1 \end{array} \right] R_2 - 3R_1$

$A^{-1} = \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & -2 & -3 & 1 \end{array} \right] R_2 \times -2$

$A^{-1} = \left[\begin{array}{cc|cc} 0 & 2 & 1 & 0 \\ 1 & 1 & 3/2 & -1/2 \end{array} \right] R_1 - 2R_2$

$A^{-1} = \left[\begin{array}{cc|cc} 1 & 0 & -2 & 1 \\ 0 & 1 & 3/2 & -1/2 \end{array} \right]$

$A^{-1} = \begin{bmatrix} -2 & 1 \\ 3/2 & -1/2 \end{bmatrix}$

Solved //

Rank Of Matrix

Q1:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 5 & 7 \end{bmatrix}$$

Sol:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 5 & 7 \end{bmatrix} \quad R_2 - 2R_1$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -2 \\ 3 & 5 & 7 \end{bmatrix} \quad R_3 - 3R_1$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -2 \\ 0 & -1 & -2 \end{bmatrix} \quad R_3 - R_2$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{bmatrix}$$

Rank of matrix = 2

Solved

Q2:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$$

Sol: $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix} \quad R_2 - 2R_1$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix} \quad R_3 - 3R_1$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

RANK OF MATRIX = 1

solved //

$$\text{Q3. } B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 4 & 7 \end{bmatrix}$$

$$\text{Sol: } B = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 4 & 7 \end{bmatrix} \quad R_2 \rightarrow R_2 - 2R_1$$

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -1 \\ 3 & 4 & 7 \end{bmatrix} \quad R_3 \rightarrow R_3 - 3R_1$$

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -1 \\ 0 & -2 & -2 \end{bmatrix} \quad R_3 \rightarrow R_3 - 2R_2$$

$$B = \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

Rank of matrix = 2
solved

Solution Of System Of Linear Equation

Q1:

$$x - y + 2z = 3$$

$$x + 2y + 3z = 5$$

$$3x - 4y - 5z = -13$$

Sol:

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & 3 \\ 1 & 2 & 3 & 5 \\ 3 & -4 & -5 & -13 \end{array} \right] \begin{array}{l} x \\ y \\ z \end{array} = \begin{array}{l} 3 \\ 5 \\ -13 \end{array}$$

$$= \left[\begin{array}{ccc|c} 1 & -1 & 2 & 3 \\ 1 & 2 & 3 & 5 \\ 3 & -4 & -5 & -13 \end{array} \right] \begin{array}{l} R_2 - R_1 \\ R_3 - 3R_1 \end{array}$$

$$= \left[\begin{array}{ccc|c} 1 & -1 & 2 & 3 \\ 0 & 3 & 1 & 2 \\ 0 & -1 & -11 & -22 \end{array} \right] 3R_3 + R_2$$

$$= \left[\begin{array}{ccc|c} 1 & -1 & 2 & 3 \\ 0 & 3 & 1 & 2 \\ 0 & 0 & -32 & -64 \end{array} \right]$$

$$= \left[\begin{array}{ccc|c} 1 & -1 & 2 & 3 \\ 0 & 3 & 1 & 2 \\ 0 & 0 & -32 & -64 \end{array} \right] \begin{array}{l} x \\ y \\ z \end{array} = \begin{array}{l} 3 \\ 2 \\ -64 \end{array}$$

$$= x - y + 2z = 3 \rightarrow 1$$

$$3y + z = 2 \rightarrow 2$$

$$-32z = -64 \rightarrow 3$$

from equation 3

$$-32z = -64$$

$$z = 64/32 \Rightarrow z = 2$$

Put value of z in equation (2)

$$3y + z = 2$$

$$3y + 2 = 2$$

$$3y = 0 \Rightarrow y = 0$$

Put the value of y and z in equation (1)

$$x - y + 2z = 3$$

$$x - 0 + 2(2) = 3$$

$$x + 4 = 3$$

$$x = 3 - 4 \Rightarrow x = -1$$

Values are

$$x = -1$$

$$y = 0$$

$$z = 2$$

Proved //

Q.2:

$$x + y + z = 6$$

$$x - y + z = 2$$

$$2x + y - z = 1$$

Sol:

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 2 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 2 & 1 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 6 \\ 2 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 & 6 \\ 1 & -1 & 1 & 2 \\ 2 & 1 & -1 & 1 \end{bmatrix} \quad R_2 - R_1$$

$$= \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & -2 & 0 & -4 \\ 2 & 1 & -1 & -1 \end{bmatrix} \quad R_3 - 2R_1$$

$$= \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & -2 & 0 & -4 \\ 0 & -1 & -3 & -11 \end{bmatrix} \quad 2R_3 - R_2$$

$$= \begin{bmatrix} 1 & 1 & 1 & 6 \\ 0 & -2 & 0 & -4 \\ 0 & 0 & -6 & -18 \end{bmatrix}$$

$$= \left[\begin{array}{ccc|c} 1 & 1 & 1 & 7 \\ 0 & -2 & 0 & 9 \\ 0 & 0 & -6 & 2 \end{array} \right] \left[\begin{array}{c} x \\ y \\ z \end{array} \right] = \left[\begin{array}{c} 6 \\ -4 \\ -18 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} x + y + z & = & 6 \\ -2y & = & -4 \\ -6z & = & -18 \end{array} \right]$$

$$z = 3$$

$$y = 2$$

$$x = 1$$

Values are 1, 2, 3

Solved

Q3:

$$x + 2y - z = 3$$

$$2x + 4y - 2z = 6$$

$$3x + 6y - 3z = 9$$

Sol:

$$\begin{bmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \\ 3 & 6 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 & -1 & | & 3 \\ 2 & 4 & -2 & | & 6 \\ 3 & 6 & -3 & | & 9 \end{bmatrix} \quad R_2 - 2R_1$$

$$= \begin{bmatrix} 1 & 2 & -1 & | & 3 \\ 0 & 0 & 0 & | & 0 \\ 3 & 6 & -3 & | & 9 \end{bmatrix} \quad R_3 - 3R_1$$

$$= \begin{bmatrix} 1 & 2 & -1 & | & 3 \\ 0 & 0 & 0 & | & 0 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 2 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix}$$

$$x + 2y - z = 3$$

here y and z are zero so we suppose that
 $y = s, z = t$ (s & t are any real number)

$$x + 2s - t = 3$$

$$x = 3 - 2s + t$$

Values are $x = 3 - 2s + t, y = s, z = t$

Solved

Consistency Of System of Linear Equations

Q1:

$$x + y + 2z = 3$$

$$x + 2y + 3z = 6$$

$$x + 3y + 5z = 9$$

Sol:

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 3 \\ 1 & 2 & 3 & 6 \\ 1 & 3 & 5 & 9 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 3 \\ 1 & 2 & 3 & 6 \\ 1 & 3 & 5 & 9 \end{array} \right] \begin{array}{l} \\ R_2 - R_1 \\ R_3 - R_1 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 3 \\ 0 & 1 & 1 & 3 \\ 0 & 2 & 3 & 6 \end{array} \right] \begin{array}{l} \\ \\ R_3 - 2R_2 \end{array}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 3 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 1 & 0 \end{array} \right]$$

$$\text{rank} = 2$$

infinite Solution

$$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 3 \\ 1 & 1 & 2 & 3 \\ 0 & 0 & 0 & 0 \end{array} \right] \begin{array}{c} x \\ y \\ z \end{array} = \begin{array}{c} 3 \\ 3 \\ 0 \end{array}$$

$$x + y + 2z = 3$$

$$x + y + 2z = 3$$

Values are -3, 6, 0 solved

Q2:

$$x + 2y + 3z = 1$$

$$2x + 4y + 6z = 5$$

$$x + y + z = 2$$

Sol:

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 5 \\ 2 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 2 & 3 & | & 1 \\ 2 & 4 & 6 & | & 5 \\ 1 & 1 & 1 & | & 2 \end{bmatrix} \quad R_2 - 2R_1$$

$$= \begin{bmatrix} 1 & 2 & 3 & | & 1 \\ 0 & 0 & 0 & | & 3 \\ 1 & 1 & 1 & | & 2 \end{bmatrix} \quad R_3 - R_1$$

$$= \begin{bmatrix} 1 & 2 & 3 & | & 1 \\ 0 & -1 & -2 & | & 1 \\ 0 & 0 & 0 & | & 3 \end{bmatrix} \quad R_3 \leftrightarrow R_2$$

Rank = 3

unique solution?

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -2 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}$$

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$$x + 2y + 3z = 1$$

$$-y - 2z = 1$$

$$0 = 3$$

System is inconsistent b/c $P(A) \neq P(A|B)$ that's why

System has no solution

solved

Q3:

$$x + y + z = 1$$

$$2x + y + 2z = 2$$

$$3x + 2y + 3z = 5$$

Sol:

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 2 \\ 3 & 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 2 \\ 3 & 2 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 5 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 & | & 1 \\ 2 & 1 & 2 & | & 2 \\ 3 & 2 & 3 & | & 5 \end{bmatrix} \begin{array}{l} \\ R_2 - 2R_1 \\ R_3 - 3R_1 \end{array}$$

$$= \begin{bmatrix} 1 & 1 & 1 & | & 1 \\ 0 & -1 & 0 & | & 0 \\ 0 & -1 & 0 & | & 2 \end{bmatrix} \begin{array}{l} \\ \\ R_3 - R_2 \end{array}$$

$$= \begin{bmatrix} 1 & 1 & 1 & | & 1 \\ 0 & -1 & 0 & | & 0 \\ 0 & 0 & 0 & | & 2 \end{bmatrix}$$

System has no consistent and has no solution.

$$P(A) \neq P(A|B)$$

Solved

Eigen Values & Eigen Vectors

Q1:

$$A = \begin{bmatrix} 5 & 4 & 2 \\ 4 & 5 & 2 \\ 2 & 2 & 2 \end{bmatrix}$$

Sol:

$$|A - \lambda I|$$
$$= \begin{vmatrix} 5-\lambda & 4 & 2 \\ 4 & 5-\lambda & 2 \\ 2 & 2 & 2-\lambda \end{vmatrix}$$

$$= (5-\lambda)[(5-\lambda)(2-\lambda) - 4] - 4[(4)(2-\lambda) - 4] + 2[8 - (5-\lambda)(5-\lambda)]$$

$$= (5-\lambda)[10 - 5\lambda - 2\lambda + \lambda^2 - 4] - 4[8 - 4\lambda - 4] + 2[8 - 10 + 2\lambda]$$

$$= (5-\lambda)(\lambda^2 - 7\lambda + 6) - 4(-4\lambda + 4) + 2(2\lambda - 2)$$

$$= (5-\lambda)(\lambda^2 - 7\lambda + 6 + 16\lambda - 16 + 4\lambda - 4)$$

$$= (5-\lambda)(\lambda^2 + 13\lambda - 14) \text{ or}$$

$$= (5-\lambda)(\lambda^2 - 7\lambda + 6) + 20\lambda - 20$$

$$= (5\lambda^2 - 35\lambda + 30 - \lambda^3 + 7\lambda^2 - 6\lambda) + 20\lambda - 20$$

$$= -\lambda^3 + 12\lambda^2 - 41\lambda + 30 + 20\lambda - 20$$

$$= -\lambda^3 + 12\lambda^2 - 21\lambda + 10$$

$$\text{if } \lambda = 1$$

$$-1 + 12 - 21 + 10 = 0$$

So, $(\lambda - 1)$ is a factor.

Dividing the polynomial by $(\lambda - 1)$

$$= (\lambda - 1)(\lambda^2 - 11\lambda + 10) = 0$$

$$= (\lambda - 1)(\lambda^2 - 10)(\lambda - 1) = 0$$

$$\lambda = 1, 1, 10$$

Eigen Values are: 1, 1, 10
For Eigen Vectors: $\lambda = 1$

$$\begin{bmatrix} 5-1 & 4 & 2 \\ 4 & 5-1 & 2 \\ 2 & 2 & 2-1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 4 & 4 & 2 \\ 4 & 4 & 2 \\ 2 & 2 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$4x + 4y + 2z = 0$$

$$4x + 4y + 2z = 0$$

$$2x + 2y + z = 0 \Rightarrow z = -2x - 2y$$

for $x = 1, y = 0$ then $z = -2$

for $x = 0, y = 1$ then $z = -2$

$$V_1 = \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}, V_2 = \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix}$$

For $\lambda = 10$

$$\begin{bmatrix} 5-10 & 4 & 2 \\ 4 & 5-10 & 2 \\ 2 & 2 & 2-10 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -5 & 4 & 2 \\ 4 & -5 & 2 \\ 2 & 2 & -8 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$-5x + 4y + 2z = 0$$

$$4x - 5y + 2z = 0$$

$$2x + 2y - 8z = 0$$

By using Cross multiplication of first two equations

$$\frac{x}{\text{determinant} \begin{vmatrix} 4 & 2 \\ -5 & 2 \end{vmatrix}} = \frac{-y}{\text{determinant} \begin{vmatrix} -5 & 2 \\ 4 & 2 \end{vmatrix}} = \frac{z}{\text{determinant} \begin{vmatrix} -5 & 4 \\ 4 & -5 \end{vmatrix}}$$

$$x = 18$$

$$y = 18$$

$$z = 8$$

divided by 8 we get

$$x = 2$$

$$y = 2$$

$$z = 1$$

Eigen Vectors are

$$u_1 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$$

$$u_2 = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$$

$$u_3 = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}$$

Solved

Q2:

$$A = \begin{bmatrix} 2 & 0 & 1 \\ -1 & 4 & -1 \\ -1 & 2 & 0 \end{bmatrix}$$

Sol:

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 2-\lambda & 0 & 1 \\ -1 & 4-\lambda & -1 \\ -1 & 2 & 0-\lambda \end{vmatrix} = 0$$

$$(2-\lambda) [(4-\lambda)(-\lambda) - (-1)(-1)] + 1 [(-1)(-\lambda) - (-1)(-2)]$$

$$(2-\lambda)(\lambda^2 - 4\lambda + 2) + (2-\lambda)$$

$$(2-\lambda)(\lambda^2 - 4\lambda + 2 + 1)$$

$$(\lambda - 2)(\lambda^2 - 3\lambda - \lambda + 3)$$

$$(\lambda - 2)(\lambda - 3)(\lambda - 1)$$

$$\lambda = 2$$

$$\lambda = 3$$

$$\lambda = 1$$

Eigen Values are

For Eigen Vectors

$$\lambda = 1$$

$$\begin{vmatrix} 2-\lambda & 0 & 1 \\ -1 & 4-\lambda & -1 \\ -1 & 2 & -\lambda \end{vmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\begin{vmatrix} 1 & 0 & 1 \\ -1 & 3 & -1 \\ -1 & 2 & -1 \end{vmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$x_1 + x_3 = 0$$

$$-x_1 + 3x_2 - x_3 = 0$$

$$-x_1 + 2x_2 - x_3 = 0$$

$$x_1 = -x_3$$

$$x_2 = 0$$

$$x_3 = 0$$

Let suppose

$$x_1 = k$$

$$x_1 = x_3$$

$$k = -x_3$$

$$x_3 = -k$$

$$x_2 = 0$$

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

For $\lambda = 2$

$$\begin{bmatrix} 0 & 0 & 1 \\ -1 & 2 & -1 \\ -1 & 2 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$x_3 = 0$$

$$-x_1 + 2x_2 - x_3 = 0$$

$$-x_1 + 2x_2 - 2x_3 = 0$$

$$x_1 = 2x_2$$

$$x_2 = -\frac{1}{2} \cdot 0$$

Let suppose

$$x_1 = k$$

$$x_2 = -\frac{1}{2}$$

$$v_2 = \begin{bmatrix} 1 \\ -\frac{1}{2} \\ 0 \end{bmatrix}$$

For $\lambda = 3$

$$\begin{bmatrix} -1 & 0 & 1 \\ -1 & 1 & -1 \\ -1 & 2 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$-x_1 + x_3 = 0$$

$$-x_1 + x_2 - x_3 = 0$$

$$-x_1 + 2x_2 - 3x_3 = 0$$

$$x_1 = x_3$$

$$x_2 = +2x_3$$

Let suppose

$$x_1 = v$$

$$x_3 = -v$$

$$x_2 = 2v$$

$$v_3 = \begin{bmatrix} v \\ 2v \\ -v \end{bmatrix}$$

Eigenvectors are

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \begin{bmatrix} 1 \\ -\frac{1}{2} \\ 0 \end{bmatrix} \begin{bmatrix} v \\ 2v \\ -v \end{bmatrix}$$

Solved

Q3:

$$A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$$

Sol:

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 1-\lambda & 4 \\ 2 & 3-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)(3-\lambda) - 8 = 0$$

$$3 - \lambda - 3\lambda + \lambda^2 - 8 = 0$$

$$\lambda^2 - 4\lambda - 5 = 0$$

$$\lambda^2 - 5\lambda + \lambda - 5 = 0$$

$$\lambda(\lambda - 5) + 1(\lambda - 5) = 0$$

$$(\lambda + 1)(\lambda - 5) = 0$$

$$\lambda = -1$$

$$\lambda = 5$$

Eigen Values are -1, 5
for Eigen Vector:

$$\lambda = -1$$

$$\begin{vmatrix} 1+1 & 4 \\ 2 & 3+1 \end{vmatrix} = 0$$

$$\begin{vmatrix} 2 & 4 \\ 2 & 4 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} 2 & 4 \\ 2 & 4 \end{vmatrix} \begin{vmatrix} x_1 \\ x_2 \end{vmatrix} = 0$$

$$2x_1 + 4x_2 = 0$$

$$2x_1 + 4x_2 = 0$$

$$x_1 = -2x_2$$

$$x_1 = k$$

$$x_2 = -k/2$$

$$x_1 = \begin{bmatrix} k \\ -k/2 \end{bmatrix}$$

For $\lambda = 5$

$$\begin{vmatrix} 1-\lambda & 4 \\ 2 & 3-\lambda \end{vmatrix} \begin{vmatrix} x_1 \\ x_2 \end{vmatrix} = 0$$

$$\begin{vmatrix} 1-5 & 4 \\ 2 & 3-5 \end{vmatrix} \begin{vmatrix} x_1 \\ x_2 \end{vmatrix} = 0$$

$$\begin{vmatrix} -4 & 4 \\ 2 & -2 \end{vmatrix} \begin{vmatrix} x_1 \\ x_2 \end{vmatrix} = 0$$

$$-4x_1 + 4x_2 = 0$$

$$2x_1 - 2x_2 = 0$$

Let Suppose $x_1 = t$

$$x_2 = t$$

$$x_2 = \begin{vmatrix} t \\ t \end{vmatrix}$$

Vectors are

$$\begin{vmatrix} x_1 \\ x_2 \end{vmatrix} = \begin{vmatrix} K \\ -K/2 \end{vmatrix}, \begin{vmatrix} t \\ t \end{vmatrix}$$

solved

Linear Dependence & Independence Of Vectors

Q1: Examine the following vectors for linear dependence and find the relation if it exists.

$$x_1 = (1, 2, 4), x_2 = (2, -1, 3), x_3 = (0, 1, 2), x_4 = (-3, 7, 2)$$

Sol:

$$\lambda_1 x_1 + \lambda_2 x_2 + \lambda_3 x_3 + \lambda_4 x_4 = 0$$

$$\lambda_1(1, 2, 4) + \lambda_2(2, -1, 3) + \lambda_3(0, 1, 2) + \lambda_4(-3, 7, 2) = 0$$

$$\lambda_1 + 2\lambda_2 + 0\lambda_3 - 3\lambda_4 = 0$$

$$2\lambda_1 - \lambda_2 + \lambda_3 + 7\lambda_4 = 0$$

$$4\lambda_1 + 3\lambda_2 + 2\lambda_3 + 2\lambda_4 = 0$$

$$\begin{bmatrix} 1 & 2 & 0 & -3 \\ 2 & -1 & 1 & 7 \\ 4 & 3 & 2 & 2 \end{bmatrix} \quad R_2 - 2R_1$$

$$\begin{bmatrix} 1 & 2 & 0 & -3 \\ 0 & -5 & 1 & 13 \\ 4 & 3 & 2 & 2 \end{bmatrix} \quad R_3 - 4R_1$$

$$\begin{bmatrix} 1 & 2 & 0 & -3 \\ 0 & -5 & 1 & 13 \\ 0 & -5 & 2 & 14 \end{bmatrix} \quad R_3 - R_2$$

$$\begin{bmatrix} 1 & 2 & 0 & -3 \\ 0 & -5 & 1 & 13 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 0 & -3 \\ 0 & -5 & 1 & 13 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \\ \lambda_4 \end{bmatrix} = 0$$

$$\lambda_1 + 2\lambda_2 - 3\lambda_4 = 0$$

$$-5\lambda_2 + \lambda_3 + 13\lambda_4 = 0$$

$$\lambda_3 + \lambda_4 = 0$$

$$\lambda_3 = -\lambda_4$$

$$\text{let } \lambda_3 = t$$

$$\lambda_4 = -t$$

$$\lambda_2 = -12/5 t$$

$$\lambda_1 = 9/5$$

$$\begin{bmatrix} \lambda_1 \\ \lambda_2 \\ \lambda_3 \\ \lambda_4 \end{bmatrix} = \begin{bmatrix} 9/5 \\ -12/5 \\ t \\ -t \end{bmatrix}$$

$$\lambda_1(1, 2, 4) + \lambda_2(2, -1, 3) + \lambda_3(0, 1, 2) + \lambda_4(3, 7, 2) = 0$$

$$9/5(1, 2, 4) - 12/5(2, -1, 3) + t(0, 1, 2) - (-3, 7, 2) = 0$$

$$1) \quad 9/5 - 12/5 + 3 = 0$$

$$2) \quad 2 \times 9/5 + 12/5 + 1 - 7 = 0$$

$$3) \quad 4 \times 9/5 - 36/5 + 2 - 2 = 0$$

Relation is dependent

solved //

Q2:

Show that the following vectors are linearly dependent by using rank method.

$$X = [1, 2, -3, 4], Y = [3, -1, 2, 1], Z = [1, -5, 8, -7]$$

Sol:

$$A = \begin{bmatrix} 1 & 2 & -3 & 4 \\ 3 & -1 & 2 & 1 \\ 1 & -5 & 8 & -7 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 2 & -3 & 4 \\ 3 & -1 & 2 & 1 \\ 1 & -5 & 8 & -7 \end{bmatrix} \begin{array}{l} R_2 - 3R_1 \\ R_3 - R_1 \end{array}$$

$$A = \begin{bmatrix} 1 & 2 & -3 & 4 \\ 0 & -7 & 11 & -11 \\ 0 & -7 & 11 & -11 \end{bmatrix} \begin{array}{l} \\ R_3 - R_2 \end{array}$$

$$A = \begin{bmatrix} 1 & 2 & -3 & 4 \\ 0 & -7 & 11 & -11 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Rank : 2

Vectors are linearly dependent

Solved

Q3:

Show whether the vectors $u = [1, 1, 2]$, $v = [2, 3, 1]$ and $w = [4, 5, 5]$ are linearly dependent or independent using the rank (matrix) method.

Sol:

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 3 & 1 \\ 4 & 5 & 5 \end{bmatrix}$$

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 3 & 1 \\ 4 & 5 & 5 \end{bmatrix} \begin{array}{l} R_2 - 2R_1 \\ R_3 - 4R_1 \end{array}$$

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & -3 \\ 0 & 1 & -3 \end{bmatrix} \begin{array}{l} R_3 - R_2 \end{array}$$

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\text{Rank} = 2$$

Vectors are linearly dependent.

solved

Cayley Hamilton Theorem

Q1:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Sol:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Find the characteristic Equation:

$$|A - \lambda I| = 0$$

$$= \begin{vmatrix} 1-\lambda & 2 \\ 3 & 4-\lambda \end{vmatrix} = 0$$

$$= \begin{vmatrix} 1-\lambda & 2 \\ 3 & 4-\lambda \end{vmatrix} = 0 \Rightarrow (1-\lambda)(4-\lambda) - 6$$

$$= \lambda^2 - 5\lambda - 2 = 0$$

Verify the Theorem:

$$A^2 - 5A - 2I = 0$$

$$A^2 = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = A^2 = \begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix}$$

$$A^2 - 5A - 2I = \begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix} - \begin{bmatrix} 5 & 10 \\ 15 & 20 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

$$A^2 - 5A - 2I = 0$$

verified

Q2: Finding A^{-1} using Cayley-Hamilton

$$A = \begin{bmatrix} 2 & 1 \\ 3 & 5 \end{bmatrix}$$

Sol:

$$A = \begin{bmatrix} 2 & 1 \\ 3 & 5 \end{bmatrix}$$

For characteristic equation

$$|A - \lambda I| = \begin{vmatrix} 2-\lambda & 1 \\ 3 & 5-\lambda \end{vmatrix} = \lambda^2 - 7\lambda + 7 = 0$$

Apply theorem

$$A^2 - 7A + 7I = 0$$

Find Inverse:

$$\text{Multiply by } A^{-1} : A - 7I + 7A^{-1} = 0$$

$$7A^{-1} = 7I - A$$

$$A^{-1} = \frac{1}{7}(7I - A)$$

$$A^{-1} = \frac{1}{7} \left(\begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix} - \begin{bmatrix} 2 & 1 \\ 3 & 5 \end{bmatrix} \right)$$

$$= \frac{1}{7} \begin{bmatrix} 5 & -1 \\ -3 & 2 \end{bmatrix}$$

$$A^{-1} = \frac{1}{7} \begin{bmatrix} 5 & -1 \\ -3 & 2 \end{bmatrix}$$

Verified

Q3. Cayley-Hamilton Verification

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$$

Sol:

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$$

Find the characteristic equation

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 1-\lambda & 0 & 2 \\ 0 & 2-\lambda & 1 \\ 2 & 0 & 3-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)[(2-\lambda)(3-\lambda)-0] - 0[(0)(3-\lambda)-2] + 2[(0)-(2-\lambda)(2)]$$

$$(1-\lambda)[6-2\lambda-3\lambda+\lambda^2-0] - 0[0-2] + 2[0-4+2\lambda]$$

$$(1-\lambda)[(2-\lambda)(3-\lambda)-0] - 0 + 2[0-(2-\lambda)(2)] = 0$$

$$(1-\lambda)(2-\lambda)(3-\lambda) - 4(2-\lambda) = 0$$

factor out $(2-\lambda)$

$$(2-\lambda)[(1-\lambda)(3-\lambda)-4] = 0$$

$$(2-\lambda)[\lambda^2 - 4\lambda + 3 - 4] = 0$$

$$(2-\lambda)(\lambda^2 - 4\lambda - 1) = 0$$

$$-\lambda^3 + 6\lambda^2 - 7\lambda - 2 = 0$$

characteristic equation : $\lambda^3 - 6\lambda^2 + 7\lambda + 2 = 0$

Apply Cayley-Hamilton theorem:

$$A^3 - 6A^2 + 7A + 2I = 0$$

$$A^2 = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix} = \begin{bmatrix} 5 & 0 & 8 \\ 2 & 4 & 5 \\ 8 & 0 & 13 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 5 & 0 & 8 \\ 2 & 4 & 5 \\ 8 & 0 & 13 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix} = \begin{bmatrix} 21 & 0 & 34 \\ 12 & 8 & 23 \\ 34 & 0 & 55 \end{bmatrix}$$

Find Substitution

A^3, A^2, A , and I into the theorem

$$= \begin{bmatrix} 21 & 0 & 34 \\ 12 & 8 & 23 \\ 34 & 0 & 55 \end{bmatrix} - 6 \begin{bmatrix} 5 & 0 & 8 \\ 2 & 4 & 5 \\ 8 & 0 & 13 \end{bmatrix} + 7 \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix} + 2 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A^3 - 6A^2 + 7A + 2I = 0$$

verified

Diagonalization

Q1:

Find a matrix 'P' which diagonalised the following matrix A. Verify $P^{-1}AP = D$

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

Sol:

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

Characteristic equation of matrix A.

As we know

$$|A - \lambda I| = 0$$
$$\begin{vmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{vmatrix} = 0$$

$$\begin{vmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{vmatrix} = 0$$

$$(4-\lambda)(3-\lambda) - 2 = 0$$

$$\lambda = 2, \lambda = 5$$

Eigen values are 2, 5.

When $\lambda = 5$

$$\begin{vmatrix} 4-5 & 1 \\ 2 & 3-5 \end{vmatrix} = 0 \Rightarrow \begin{vmatrix} -1 & 1 \\ 2 & -2 \end{vmatrix} \begin{vmatrix} x_1 \\ x_2 \end{vmatrix} = 0$$

$$-x_1 + x_2 = 0$$

$$2x_1 - 2x_2 = 0$$

$$x_1 = x_2$$

Let Suppose $x_1 = k$ then $x_2 = k$

$$x_2 = \begin{vmatrix} k \\ k \end{vmatrix} \text{ or } \begin{vmatrix} 1 \\ 1 \end{vmatrix}$$

when $\lambda = 2$

$$\begin{vmatrix} 4-2 & 1 \\ 2 & 3-2 \end{vmatrix} = 0 \Rightarrow \begin{vmatrix} 2 & 1 \\ 2 & 1 \end{vmatrix} \begin{vmatrix} x_1 \\ x_2 \end{vmatrix} = 0$$

$$2x_1 + x_2 = 0$$

$$2x_1 + x_2 = 0$$

$$x_2 = -2x_1$$

Suppose $x_1 = k$ then $x_2 = -2k$

$$x_1 = \begin{vmatrix} k \\ -2k \end{vmatrix} \text{ or } \begin{vmatrix} 1 \\ -2 \end{vmatrix}$$

modal matrix $\Rightarrow P = \begin{bmatrix} 1 & 1 \\ -2 & 1 \end{bmatrix} \Rightarrow P^{-1} = \frac{1}{3} \begin{vmatrix} 1 & -1 \\ 2 & 1 \end{vmatrix}$

$$P^{-1} = \frac{1}{D} \times \text{adj}$$

for determinant

$$D = 3$$

and adjacent matrix = $\begin{vmatrix} 3 & 3 \\ -6 & 3 \end{vmatrix}$

$$P^{-1} = \frac{1}{3} \begin{vmatrix} 3 & 3 \\ -6 & 3 \end{vmatrix}$$

$$P^{-1} = \frac{1}{3} \begin{vmatrix} 1 & 1 \\ -2 & 1 \end{vmatrix}$$

for diagonalization

$$D = P^{-1}AP = \frac{1}{3} \begin{vmatrix} 1 & 1 \\ -2 & 1 \end{vmatrix} \begin{vmatrix} 4 & 1 \\ 2 & 3 \end{vmatrix} \begin{vmatrix} 1 & 1 \\ -2 & 1 \end{vmatrix}$$

$$= \frac{1}{3} \begin{vmatrix} 1 & 1 \\ -2 & 1 \end{vmatrix} \begin{vmatrix} 2 & 5 \\ -4 & 5 \end{vmatrix}$$

$$= \frac{1}{3} \begin{vmatrix} 6 & 0 \\ -6 & 15 \end{vmatrix}$$

$$D = \begin{vmatrix} 2 & 0 \\ 0 & 5 \end{vmatrix}$$

verified

Q2:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & -1 \\ 0 & -1 & 3 \end{bmatrix}$$

Sol:

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & -1 \\ 0 & -1 & 3 \end{bmatrix}$$

Show that

$$D = P^{-1}AP$$

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 1-\lambda & 0 & 0 \\ 0 & 3-\lambda & -1 \\ 0 & -1 & 3-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)[(3-\lambda)(3-\lambda) - 1] = 0 + 0$$

$$(\lambda - 1)[\lambda^2 - 6\lambda + 8]$$

$$\lambda = 1 [(\lambda^2 - 3\lambda - 3\lambda + 8)]$$

$$\lambda = 1 [\lambda^2 - 6\lambda + 8]$$

$$\lambda_1 = 1$$

$$\lambda_2 = 2$$

$$\lambda_3 = 4$$

Eigen values are 1, 2, 4

For $\lambda = 1$

$$\begin{vmatrix} 1-1 & 0 & 0 \\ 0 & 3-1 & -1 \\ 0 & -1 & 3-1 \end{vmatrix} \Rightarrow \begin{bmatrix} 0 & 0 & 0 \\ 0 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$2x_1 + x_3 = 0$$

$$-x_2 + 2x_3 = 0$$

$$2x_2 = x_3$$

$$x_2 = x_3/2$$

$$-x_2 + 2x_3 = 0$$

$$x_3 = 0$$

$$x_1 = 1$$

$$x_2 = 0$$

$$x_3 = 0$$

$$V_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{For } \lambda_2 = 2$$

$$\begin{bmatrix} 1-2 & 0 & 0 \\ 0 & 3-2 & -1 \\ 0 & -1 & 3-2 \end{bmatrix} = 0$$

$$\begin{bmatrix} -1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$-x_1 = 0$$

$$x_2 - x_3 = 0$$

$$-x_2 + x_3 = 0$$

$$x_1 = 0$$

$$x_2 = x_3$$

Let suppose

$$x_2 = 1$$

$$x_3 = 1$$

$$V_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$\text{For } \lambda_3 = 4$$

$$\begin{bmatrix} 1-4 & 0 & 0 \\ 0 & 3-4 & -1 \\ 0 & -1 & 3-4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$-3x_1 = 0$$

$$-x_2 - x_3 = 0$$

$$-x_2 - x_3 = 0$$

$$x_1 = 0$$

$$-x_2 = x_3$$

$$x_2 = 1$$

$$x_3 = -1$$

$$V_3 = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

Eigen Vectors are

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

hence $P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & -1 \end{bmatrix} \Rightarrow \text{adj} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & -1 \\ 0 & -1 & 1 \end{bmatrix}$

here $D = -2$

$P^{-1} = \frac{1}{D} \times \text{adjacent}$

$P^{-1} = \frac{1}{2} \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & -1 \\ 0 & -1 & 1 \end{bmatrix}$

$P^{-1} = \frac{1}{2} \begin{bmatrix} 2 & 0 & 0 \\ 0 & -1 & -1 \\ 0 & -1 & 1 \end{bmatrix} \Rightarrow P^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 1/2 \\ 0 & 1/2 & -1/2 \end{bmatrix}$

For $D = P^{-1}AP$

$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 1/2 \\ 0 & 1/2 & -1/2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & -1 \\ 0 & -1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 1 & -1 \end{bmatrix}$

$D = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{bmatrix}$

Proved